

Let $\Gamma \curvearrowright X$. Γ acts p.d. iff $\forall x, y \in X$,
 $\exists$ nbhd $U \ni x, V \ni y$ s.t. $|\{g \in \Gamma \mid gU \cap V \neq \emptyset\}| < \infty$
 If X is locally cpt, this is equivalent to:

$\forall A, B \subseteq X$ cpt, $|\{g \in \Gamma \mid gA \cap B \neq \emptyset\}| < \infty$.
 If $\Gamma < G$, then, Γ is discrete iff it is discrete
 w.r.t. the topology on G .

Thm: Let $\Gamma < G$ be discrete, then it is closed and without
 accumulation pts.

pf: $\exists$ a nbhd U of 1 in G s.t. $U \cap \Gamma = \{1\}$.
 Let V be a nbhd of 1 s.t. $V^{-1}V \subseteq U$. For any
 distinct $\alpha, \beta \in \Gamma$, $\alpha V \cap \beta V = \emptyset$. Let $g \in \Gamma$ and
 $\alpha \in V^{-1}g \cap \Gamma$, then $V^{-1}g \cap \Gamma = \{\alpha\} \Rightarrow g = \alpha \in \Gamma$. $\square$

Thm: Let G be an l.c. second countable gp, X l.c. Haus.
 $q: G \curvearrowright X$ is transitive, then if $\Gamma < G$ and G_x is
 cpt $\forall x \in X$, then:

- 1) Γ is discrete
- 2) Γ acts p.d. on X :

pf: (1) $\Rightarrow$ (2) Let $x \in X$, $A, B \subseteq X$ cpt. Put
 $M = \{g \in G \mid gx \in A\}$, $N = \{g \in G \mid gx \in B\}$. Since
 $G \curvearrowright X$ is transitive, $A = Mx$. Take an open cover
 $M \subseteq \bigcup U_i$ s.t. $\overline{U_i}$ is cpt $\forall i$ which can exist b.c.
 G is l.c. $A \subseteq \bigcup U_i x$. Choose a finite subcover
 by cptness $A \subseteq \bigcup_{i=1}^n U_i x$. Thus $M \subseteq \bigcup_{i=1}^n U_i G_x$. Since
 each $U_i G_x$ is cpt and M is closed, M is cpt.
 Also, N is cpt. Since Γ is discrete and NM^{-1} is
 cpt, $\{g \in \Gamma \mid gA \cap B \neq \emptyset\} = NM^{-1} \cap \Gamma$ is cpt
 and discrete, thus finite.

($\Leftarrow$) Let $V \subseteq G$ be an open nbhd of 1 s.t. $\overline{V}$ is cpt.
 For $x \in X$, $\Gamma \cap V$ is included in $\{g \in \Gamma \mid gx \in \overline{V}x\}$
 which is finite by p.d. ($\{gx\}$ and $\overline{V}x$ are cpt). Hence
 take U s.t. $\Gamma \cap U = \{1\}$.

Cor

Any discrete subgroup of $SL_2(\mathbb{R})$ acts p.d. on $\mathbb{H}$.
 Conversely, and subgroup of $Aut(\mathbb{H})$ which acts p.d.
 on $\mathbb{H}$ is obtained from a discrete subgroup of $SL_2(\mathbb{R})$.

A discrete subgroup of $PSL_2(\mathbb{R})$ is called a Fuchsian group.
 From now on: Γ is Fuchsian and $Z(\Gamma) = \Gamma \cap \{\pm I\}$.

If $\iota: SL_2(\mathbb{R}) \rightarrow Aut(\mathbb{H})$ is the natural homom., then
 $\iota(\Gamma) = \Gamma / Z(\Gamma)$.

Recall: Let $\alpha \in GL_2(\mathbb{R})^+$

- 1) α is elliptic if $tr(\alpha)^2 < 4 \det(\alpha)$
- 2) α is parabolic if $tr(\alpha)^2 = 4 \det(\alpha)$
- 3) α is hyperbolic if $tr(\alpha)^2 > 4 \det(\alpha)$

Let $z \in \mathbb{H} \cup \mathbb{R} \cup \{\infty\}$, then z is elliptic, parabolic, or hyperbolic
 iff it is fixed by an elliptic, parabolic, or hyperbolic elt of Γ
 resp. A parabolic pt is also called a cusp of Γ .

If $z \in \mathbb{H} \cup \mathbb{R} \cup \{\infty\}$, $x, x' \in \mathbb{R} \cup \{\infty\}$, then

$$\Gamma_z = \Gamma \cap SL_2(\mathbb{R})_z, \quad \Gamma_{x,x'} = SL_2(\mathbb{R})_{x,x'} \cap \Gamma.$$

Prop: If z is an elliptic pt, then Γ_z is finite cyclic.

2) If $x \in \mathbb{R} \cup \{\infty\}$ is a cusp of Γ , then $\Gamma_x \subseteq SL_2(\mathbb{R})_x^{(p)}$
 and $\Gamma_x / Z(\Gamma) \cong \mathbb{Z}$. If $\sigma \in SL_2(\mathbb{R})$ s.t. $\sigma x = x$, then
 $\sigma \Gamma_x \sigma^{-1} \cdot Z(\Gamma) = \{ \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \mid n \in \mathbb{Z} \}$ ($n > 0$)

3) If $\Gamma_{x,x'} \neq Z(\Gamma)$ for $x, x' \in \mathbb{R} \cup \{\infty\}$ ($x \neq x'$), then
 $\Gamma_{x,x'} / Z(\Gamma) \cong \mathbb{Z}$ and $\exists \sigma \in SL_2(\mathbb{R})$ s.t.

$\sigma \Gamma_{x,x} \sigma^{-1} \cdot \{\pm I\} = \{\pm \begin{pmatrix} u & 0 \\ 0 & u^{-1} \end{pmatrix} \mid u \in \mathbb{R}^{\times}\}$ for some $u > 0$.

Pf: 1) Since $SL_2(\mathbb{R}) \subset H$ is transitive, $SL_2(\mathbb{R})_z$ is conjugate to $SL_2(\mathbb{R})_i = SO(2)$ which is a cyclic group. Isom. to $\mathbb{T}^1$, the circle group. Thus $\Gamma_z = \Gamma \cap SL_2(\mathbb{R})_z$ is finite + cyclic.

2) wlog, $x = \infty$, hence $\exists \gamma \in \Gamma_x$ s.t. $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \gamma$ for some $1 \neq 0$. Suppose that $\alpha \in \Gamma_x$ is neither parabolic nor scalar, then write $\alpha = \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix}$ w/ $a \neq \pm 1$. Assume $|a| < 1$. We get $\alpha^n \gamma \alpha^{-n} = \begin{pmatrix} 1 & a^{2n} \\ 0 & 1 \end{pmatrix} \in \Gamma \quad \forall n \in \mathbb{N}$, which goes to ∞ as $n \rightarrow \infty$, contradicting discreteness of Γ . Therefore:

$\Gamma_\infty \subset SL_2(\mathbb{R})_\infty^{(p)} = \{\pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mid t \in \mathbb{R}\}$. Since $SL_2(\mathbb{R})_\infty^{(p)} / \{\pm I\} \cong \mathbb{R}$, we get (2).

3) From a previous lemma: $\exists \gamma \in SL_2(\mathbb{R})$ s.t.

$$\sigma \Gamma_{z,x} \sigma^{-1} \subset \sigma SL_2(\mathbb{R})_{x,x} \sigma^{-1} = SL_2(\mathbb{R})_{\infty,\infty} = \left\{ \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \mid a \in \mathbb{R}^{\times} \right\}$$

Since discrete subgrps of $\mathbb{R}^{\times} / \{\pm 1\}$ are cyclic, we get 3).

For any $z \in H$, we define $e_z = |\Gamma_z / \mathbb{Z}(\Gamma)|$ and call it the order of z wrt Γ . If z is a cusp at Γ , we define we have z is elliptic iff $e_z > 1$ (Note: only true for $z \in H$!). If z is a cusp at Γ , define $e_z = \infty$. By the above, we have that:

1) for z elliptic, $e_z < \infty$.

2) If z is hyperbolic, it cannot be a cusp.

Cor: If $\Gamma' < \Gamma$ w/ Γ Fuchsian w/ $[\Gamma' : \Gamma] < \infty$, then the cusps of Γ' are the same as those of Γ .

Pf: Since cusps of Γ' are also cusps of Γ by definition, we only need the converse: Let x be a cusp of Γ , then since $[\Gamma_x : \Gamma_x \cap \Gamma'] \leq [\Gamma : \Gamma'] < \infty$, then by the above

then $\Gamma'_x = \Gamma_x \cap \Gamma' \neq \mathbb{Z}(\Gamma')$ so x is also a cusp of Γ' .

Fundamental Domains:

A If Γ is a Fuchsian gp. and F is a connected domain of H , F is a fundamental domain for Γ iff:

- 1) $H = \bigcup_{\gamma \in \Gamma} \gamma F$
- 2) $F = U$ w/ U open and $F^\circ = U$.
- 3) $\gamma U \cap U = \emptyset \quad \forall \gamma \in \Gamma \setminus \mathbb{Z}(\Gamma)$

Ex: Consider the set $F_x = \{z \in H \mid d(z, z_0) \leq d(z, \gamma z_0)\}$ for $\gamma \in \Gamma$.

Define the sets $F_x = \{z \in H \mid d(z, z_0) \leq d(z, \gamma z_0)\}$
 $C_x = \{z \in H \mid d(z, z_0) = d(z, \gamma z_0)\}$

Then the set $F = \bigcap_{\gamma \in \Gamma} F_x$ is a fundamental domain for Γ . Moreover:

1) Any geodesic joining two points in F is contained in F (F is convex).

2) Let $L_\gamma = F \cap \gamma F$ for $\gamma \in \Gamma \setminus \mathbb{Z}(\Gamma)$, then $L_\gamma \subseteq C_\gamma$. If $L_\gamma \neq \emptyset$ it is either 1 point or a geodesic.

3) For any cusp M , $\{\gamma \in \Gamma \mid M \cap \gamma F \neq \emptyset\} < \infty$.